Gravitation CHEAT SHEET

KEPLER'S THREE LAWS

1 orbits are ellipses, the Sun at one focus

a = semi-major axis = (closest + farthest)/2

2 equal areas in equal times: $\frac{\Delta A}{\Delta t} = \frac{L}{2m}$

so $r_p v_p = r_a v_a$: 1 AU and 3 AU → 3 × faster

simulated: 8 equal times → 8 areas of 1.36 AU^2

3 T^2/a^3 is the same for every planet: 1 in years and AU

a = 4 AU → T = 8 yr; T = 27 yr → a = 9 AU

CONSTANTS USED

$G = 6.674 \times 10^{-11}$ N m^2/kg^2 (CODATA 2022)

Earth: $M = 5.972 \times 10^{24}$ kg, R = 6,371 km

$GM = 3.986 \times 10^{14}$ m^3/s^2

Moon: 384,400 km, 27.32 days, 7.346×10^{22} kg

sidereal day 23 h 56 min 4 s = 86,164.09 s

NEWTON'S LAW OF GRAVITATION

$F = \frac{G m_1 m_2}{r^2}$ (always attracts, along the line of centres)

inverse square: $a = \frac{4\pi^2 r}{T^2}$ with $T^2 = kr^3$ gives $a = \frac{4\pi^2}{kr^2}$

third-law pairs: you pull Earth with your 589 N too

superposition: add the pulls as arrows

$T^2 = \frac{4\pi^2}{GM} r^3$ (the orbiting mass cancels)

simulated 7,000 / 14,000 km: 97 / 275 min, ratio 2.83

weigh the Sun: $M = \frac{4\pi^2 r^3}{GT^2} = 1.99 \times 10^{30}$ kg

THE MOON TEST AND THE SHELLS

Moon: r ≈ 60 R, so $a = g/60^2 = 0.00272$ m/s^2

from its orbit: $4\pi^2 r/T^2 = 0.00272$ m/s^2: it matches

in 1 min it falls 4.9 m from a straight line (simulated)

a ball pulls as if all its mass sat at its centre

inside a hollow shell: no pull at all (20,000 rings)

G ON, ABOVE AND BELOW THE GROUND

$g = \frac{GM}{R^2} = 9.82$ m/s^2 (we use 9.8)

height: $g_h = g\frac{R^2}{(R+h)^2} \approx g(1 - \frac{2h}{R})$

400 km: 8.68 exact, 8.57 by the shortcut

depth: $g_d = g(1 - \frac{d}{R})$ (uniform ball)

1 % of R up: −2 %; 1 % down: −1 %

half g at $h = (\sqrt{2} - 1)R = 2{,}639$ km

EDGE TESTS

circle: e = 0, the two foci meet, steady speed

r → ∞: F, U, V and E all rise or fall to 0

h = R: g/4 = 2.45; centre: g = 0

lowest orbit: 7.91 km/s, 84 min

Moon: g = 1.62 m/s^2, escape 2.38 km/s

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POTENTIAL ENERGY AND POTENTIAL

near the ground: $U = mgh$ (g the same all the way)

$U = -\frac{GMm}{r}$ (zero only infinitely far away)

= the area under F–r from r to ∞: 62.6 MJ per kg at R

lift by h: $\Delta U = mgh \cdot \frac{R}{R+h}$ → mgh when h is small

1,000 kg to 400 km: 3.92 GJ (mgh) vs 3.69 GJ

potential $V = -\frac{GM}{r}$: −62.6 MJ/kg at R

CHECK BEFORE YOU TRUST IT

before you trust an answer about gravity:

1 is every distance measured from the centre?

2 is the energy zero far away, and negative nearer?

3 did the small body's mass cancel?

4 is the period the one the problem needs (sidereal?)

ESCAPE SPEED AND ORBITS

escape: $v = \sqrt{2GM/R} = \sqrt{2gR} = 11.2$ km/s

10 km/s straight up → 25,400 km high (simulated)

orbit: $v = \sqrt{GM/r}$; $v_{esc} = \sqrt{2} \cdot v_{orbit}$

period: $T = 2\pi\sqrt{r^3/GM}$: 400 km → 92.4 min

energy: $KE = \frac{GMm}{2r}$, $E = -\frac{GMm}{2r} = -KE$

higher: add energy, and it slows down

400 → 800 km, 1,000 kg: +1.64 GJ, 7.67 → 7.46 km/s

over one spot: above the equator, T = the sidereal day

r = 42,164 km, h ≈ 35,800 km, v = 3.07 km/s

PYTHON AND THE SIMULATION

Python: `v = math.sqrt(G * M / r)`

`T = 2 * math.pi * math.sqrt(r**3 / (G * M))`

r for a period: `(G*M*T**2 / (4*math.pi**2)) ** (1/3)`

simulate: RK4 steps of $a = -GM\,r/|r|^3$

each run checked against its formula